DATA MINING

HW 2

Anomaly Detection

Assignment specification and grading criteria

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Submission Deadline
Mon 2026-11-16 23:59

Presentation
Tue 2026-11-17, in class

OVERVIEW

Task Introduction

Each team defines its own anomaly detection application. The task starts from a real need, targets rare events that are worth detecting, and produces a result that can be evaluated.

You are NOT required to beat the state of the art.

This assignment grades experimental rigor, ablation design, and error analysis — not who has the fanciest model architecture. A simple Primary Method that loses to a strong reference, but is evaluated on a clean shared split, ablated carefully, and analyzed with concrete failure cases, scores higher than a complex model with shallow analysis.

Permitted subjects are unrestricted. Representative examples:

– Credit-card fraud · Machine vibration faults · Network intrusion · Surface defect inspection · Abnormal ECG beats · Server log failures

What is assessed

– Model complexity is not assessed. Assessment covers the clarity of the anomaly definition, the fairness of the comparison, the depth of ablations and error analysis, and the verifiability of the result.

– Creative, distinctive projects also earn higher scores.

OVERVIEW

Scope of the Assignment

The detection task is defined by the team. The data, the method, and the evaluation setup are documented in the report.

You decide

- Application scenario and data source
- What counts as an anomaly in your domain
- Data category (one of the 5 on the next slide)
- Tier 2 Primary Method (e.g., Isolation Forest, Autoencoder, One-Class SVM, LOF)
- Learning paradigm: unsupervised, semi-supervised, or supervised
- Which AI tools to use, and for what
- Evaluation metrics that reflect your domain's cost of error (at least 2 metrics)

Fixed for every group

- At least 3 methods: Tier 1 Naive Baseline, Tier 2 Primary Method, Tier 3 Strong / SOTA Reference
- Real-world data with a documented source (public or self-collected) and ground-truth anomaly labels for the test set
- All methods share the same data split and metrics
- One-class / reconstruction models are trained on normal data only; a small validation set of labeled anomalies may be used strictly for threshold tuning (τ) and model selection (never for fitting parameters)
- Findings are reported as a short ACL-style paper

OVERVIEW

Application Selection: The 5 Data Categories

Your dataset must belong to one of these categories; discuss its specific challenge in your paper.

Category and key challenge

– **1. Time-Series:** vibration, ECG, grid power. Temporal dependencies, windowing, and evaluation granularity (point-wise vs. event-based; point-adjustment is discouraged due to score inflation).

– **2. Tabular:** card transactions, clinical tests. High dimensionality and contextual anomalies.

– **3. Image / Video:** surface defects, CT scans. Spatial texture and background noise.

– **4. Text / Logs:** server error logs, bot posts. Unstructured vocabulary and semantic drift.

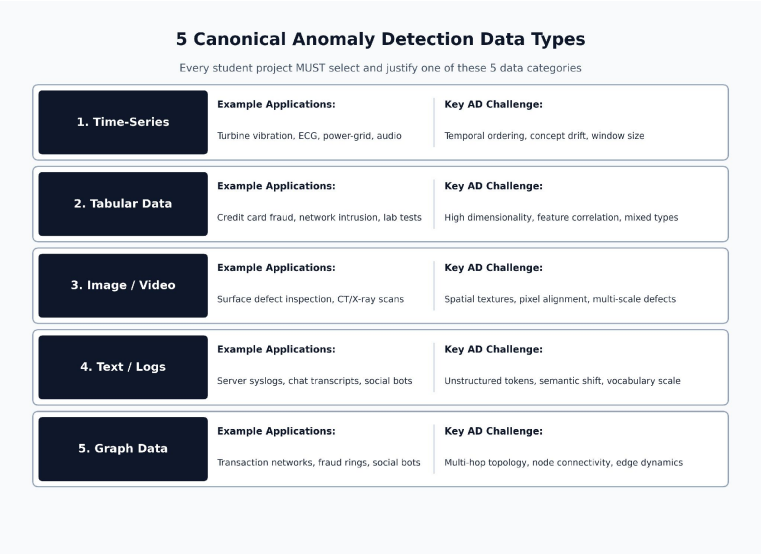
– **5. Graph:** transaction networks, fraud rings. Multi-hop relational structure.

Requirements

– State the category explicitly in §1 and §2.

– Justify why your method family suits this category.

(EXAMPLE)



OVERVIEW

The Mandatory 3-Tier Method Comparison

Compare against baselines of increasing strength; losing to Tier 3 is acceptable if you explain why.

Implement and compare at least 3 methods

– **Tier 1 · Naive Baseline:** open-ended simple heuristic (e.g., predict-all-normal / always-false, random score, single-feature threshold, or 3σ). Shows the performance floor and exposes class-imbalance pitfalls.

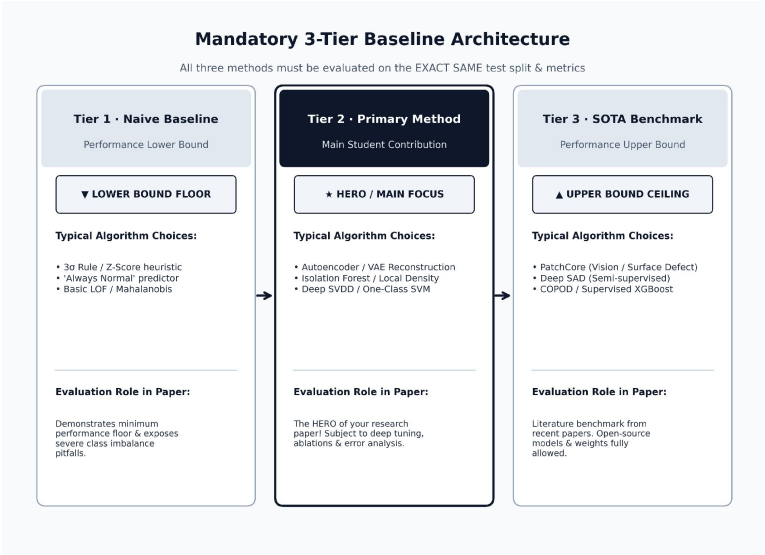
– **Tier 2 · Primary Method:** the method your team implements, tunes, and ablates in depth (e.g., Isolation Forest, Autoencoder). The focus of your paper.

– **Tier 3 · Strong / SOTA Reference:** a competitive method from the literature (e.g., PatchCore, Deep SAD, supervised XGBoost). Public packages and pretrained weights are allowed with citation.

Fair comparison

– All three tiers share the exact same raw data split, label partition, and test evaluation metrics. Method-specific preprocessing (e.g., standard scaling for Tier 2 vs. pretrained transforms for Tier 3) is permitted but must be documented.

(EXAMPLE)



PAPER REPORT

Paper Report: Write a Short ACL Format Paper

Present your work as a research paper, not as answers to a question list. The next slides guide each section.

Template

– Official ACL LaTeX template on Overleaf (search “ACL Template” on overleaf.com). Use \usepackage[preprint]{acl} (shows names and page numbers).

Length

– At most 6 pages, excluding references. Content after page 6 is not read. Keep the template's margins and font size. The Contributions section counts toward the 6 pages.

Front matter

– Title, team ID, every member's name and student ID, and the URL of the project's GitHub repository.

Figures & tables

– Numbered, captioned, and referenced in the text; axes labeled with units.

References

– Real, verifiable sources in ACL citation style via BibTeX and \citet{} / \citep{}.

(EXAMPLE)



Forecasting Rooftop PV Output with Interpretable Linear Models

Member A (ID) and Member B (ID) and Member C (ID)
Team G00, Data Mining (Fall 2026), NYCU
https://github.com/<team>/hw1

Abstract

We study hour-ahead photovoltaic (PV) output forecasting for a 5 kW rooftop plant. Using weather-forecast features and a small set of physically motivated engineered features, a ridge regression model reaches a mean absolute error of 1.16 kWh on a held-out five-month test period, outperforming persistence, single-feature regression and a random forest trained on the raw inputs. Coefficient and error analyses show that irradiance and clear-sky geometry dominate, that temperature has a negative conditional effect, and that the largest errors occur under unforecast convective storms.

1 Introduction

[Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone. [Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone.

2 Related Work

[Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone. [Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone.

3 Task and Data

3.1 Task Definition

[Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone. [Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone.

3.2 Data Collection and Preprocessing

[Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone. [Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone.

4 Exploratory Feature Analysis

[Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone. [Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone.

1

Method | CV MAE | Test MAE
Persistence | 2.53 ± 0.19 | 2.60
Single-feature OLS | 2.18 ± 0.16 | 2.08
Random forest | 1.52 ± 0.11 | 1.33
Ours | 1.36 ± 0.10 | 1.16

Table 1: Main results (kWh).

7 Results

[Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone. [Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone.

8 Analysis

[Placeholder text] We describe what is predicted, for whom, and which decision the prediction supports. Every claim made in this section is verified by a numbered experiment later in the paper, and each design choice is justified with evidence from the data rather than by intuition alone. [Place-

2

Paper report example in the official ACL template.

Paper Structure and Section Weights

Each section answers one guiding question. The paper report is scored out of 100 as below, then weighted 70% of HW2.

Section	Guiding question	Points
Abstract	Why does detecting these anomalies matter, to whom, and what do you contribute?	7
1 Introduction		
2 Related Work	What have others done, and how does your work differ?	6
3 Task Definition & Data	What counts as an anomaly, which data source, and how are ground truth labels defined?	12
4 Exploratory Feature Analysis	What do normal vs. abnormal distributions suggest before any modeling?	10
5 Method	How do raw data become an anomaly score and a decision threshold, and why this design?	13
6 Experimental Setup	Which data split, leakage prevention protocol, and metrics make the comparison fair?	10
7 Results	Does your design beat the baselines, and what do ablations prove?	18
8 Analysis	What did the model learn (interpretability), and where does it fail (FP vs. FN)?	16
9 Conclusion & Limitations	What should the reader take away, what blocks deployment, and what comes next?	5
Contributions of Each Member	Who did what, and how were AI tools used?	3
References	Real, verifiable sources in ACL citation style (outside the 6 page limit)	—
Paper Report Total		**100**

Use these sections, in this order, as the headings of your paper; section titles may be adapted to your topic.

Abstract & Introduction

Motivate the detection problem and state what your paper contributes.

7 pts
out of 100

Abstract (≤150 words)

– Application, data, the main quantitative result.

Scenario & Stakeholders

– Who uses the alerts, and what action follows an alert (freeze, shutdown, manual audit)?

Motivation

– Why this application, and what makes it technically interesting for anomaly detection?

Definition of 'Anomaly'

– Define precisely what an anomaly is in your task; give one normal and one abnormal sample.

Contributions

– 2–4 contribution statements, each verified later in §7–§8. Empirical findings count; no new architecture needed.

(EXAMPLE) Running Example: Card Fraud (ULB)

Weak motivation: “We detect fraudulent transactions using an autoencoder.”

Strong motivation: “Risk operations must flag stolen cards in real time (<50 ms) to freeze unauthorized charges while keeping false alerts under review budget; our detector drives automated block vs. challenge decisions.”

Research question: “Can semi-supervised reconstruction error isolate rare credit frauds under extreme 0.17% class imbalance without hurting legitimate user throughput?”

Concrete samples:

- Normal sample: In-store chip purchase, $34.20, local grocery, 14:30 weekday
- Anomaly sample: Rapid overseas wire, $4,850.00, unseen IP, 03:15 AM

Primary model & score: Deep Autoencoder (Tier 2); score $s(x)$ = MSE reconstruction error

Operational action: Score $s(x) > \tau$ triggers instant card freeze & SMS verification challenge

All EXAMPLE boxes show the expected form only.

Common deductions Vague motivation with no identified user or action; anomaly never defined; contributions that later results do not support.

Related Work

Position your work against existing studies, datasets, and deployed systems.

6 pts
out of 100

Data Category & Literature

– Cite at least 3 relevant sources: papers, technical reports, benchmark datasets, or deployed systems.

– For each, summarize the data, method family, paradigm, and reported result.

– Cite with BibTeX and \citet{} / \citep{} as in the ACL template.

Positioning & Method Selection

Why these 3 tiers? Which assumptions or heuristics does each tier rely on (e.g., naive prior / rule-based for Tier 1, reconstruction / density / isolation for Tier 2, deep representations / supervision for Tier 3)?

– State explicitly what your work does differently: data, setting, analysis, or user.

(EXAMPLE) A compact comparison table

Work	Data	Method	Paradigm	PR-AUC
Prior [1]	ULB card fraud	Isolation Forest	Unsup.	0.27
Prior [2]	ULB card fraud	XGBoost	Supervised	0.85
Prior [3]	ULB card fraud	Autoencoder	Semi-sup.	0.48
Ours	ULB card fraud	AE + tuned τ	Semi-sup.	0.62

[1]–[3] stand for cited papers. Results on different splits give context, not a direct comparison. Values are illustrative.

Every reference must be real and verifiable. Check each citation on the proceedings or the publisher's page. Fabricated references are penalized. Blog posts or Kaggle notebooks alone do not count as the 3 sources.

Common deductions Citations listed without any comparison; fewer than 3 sources; unverifiable or fabricated references.

Task Definition, Labels & Class Imbalance

Define what counts as an anomaly, your data source, and how extreme imbalance is handled.

12 pts
out of 100

Task & Anomaly Definition

– Define precisely what counts as normal vs. anomalous in your application domain.

– Specify the unit of detection (e.g., transaction row, sensor time-window, image patch); provide one concrete normal and one abnormal sample.

Dataset Specifications & Labels

– Source URL / citation, sample count N, feature dimension D, anomaly count, and a summary-statistics table.

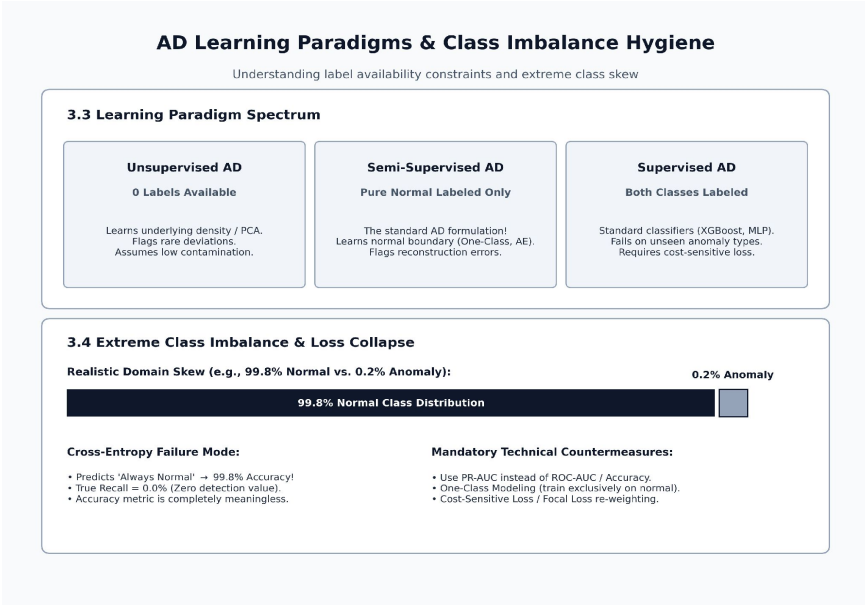
– Ground truth label acquisition (expert audit, automated rule, or injected fault); acknowledge potential label noise and paradigm (unsupervised, semi-supervised, or supervised).

Class Imbalance

– Exact ratio: e.g., 99.83% normal vs. 0.17% anomaly (1 : 577).

– Effect: why plain cross-entropy or accuracy collapses to 'always normal'; countermeasures.

(EXAMPLE)



Common deductions Undefined anomaly definition; imbalance ratio missing; accuracy used as main metric; preprocessing without justification.

Exploratory Feature Analysis

Show evidence of abnormal patterns and feature separability before building any model.

10 pts
out of 100

Distribution Contrast

– At least one figure contrasting normal vs. abnormal distributions (feature histograms, KDE, scatter plots, PCA/t-SNE projection, or sensor profiles).

Feature Separability & Association

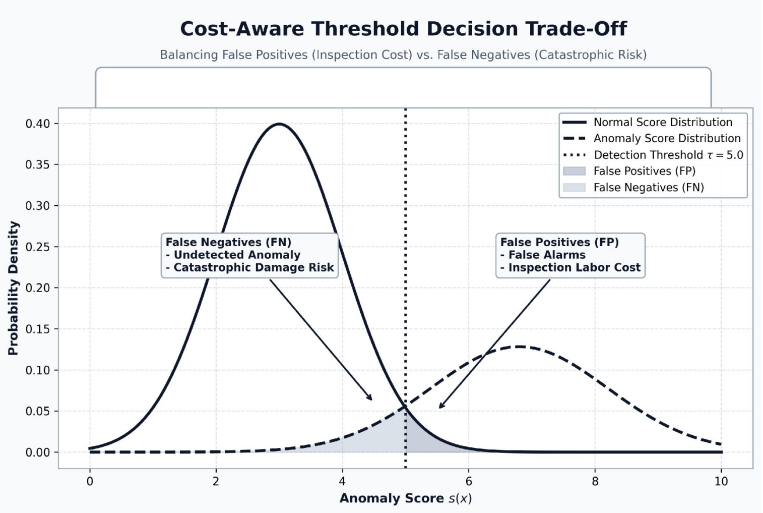
– Quantify separation (e.g., mutual information, Wasserstein/KS distance, correlation); check redundancy among candidate features.

Exploratory Hypotheses

– Explain evidence that led you to select, transform, or engineer features; state hypotheses (H1, H2, ...) to test in §7–§8.

– Strict rule: Use the training portion only, so the test set remains unseen!

(EXAMPLE)



Common deductions Figures without interpretation; statistics computed on test data; hypotheses that are never revisited.

Pipeline, Scoring & Threshold Decision

Describe how raw data become an anomaly score and an operational alert decision.

13 pts
out of 100

End-to-End Pipeline Figure

– One figure: draw complete pipeline for Tier 2 Primary Method (suggested flow: input tensor, preprocessing, core model, scoring s(x), threshold τ, output alerts; adapt to your architecture).

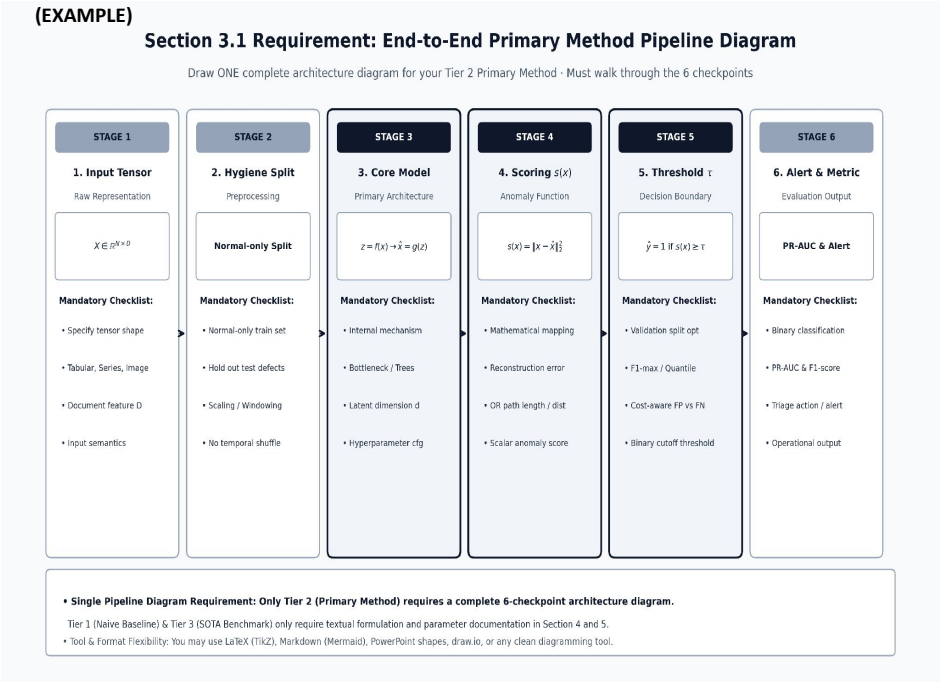
Anomaly Scoring & Threshold Decision

– Scoring function: formulate s(x) exactly (reconstruction MSE, path length, Mahalanobis).

– Threshold τ: fixed percentile, best F1 on validation split (with labeled anomalies), or review-budget rule — chosen strictly without the test set.

– FP vs. FN cost: describe operational cost of each error; state which error your τ suppresses and why.

(EXAMPLE)



Common deductions Missing pipeline figure; score not formulated; τ chosen on test set or not reported; no discussion of FP vs. FN cost.

Validation Protocol & Configurations

Estimate how well the methods generalize to unseen data, under identical conditions.

10 pts
out of 100

Split & Leakage Prevention

– Train / val / test ratios; one shared split (fixed indices or seed) for all 3 methods.

– Normal-only training for one-class/reconstruction models; validation set may include labeled anomalies strictly for tuning τ and hyperparameters (never for training weights); chronological split for time series.

– Preprocessing fit on training portion only; hold out test set and evaluate once.

Configurations & Hardware

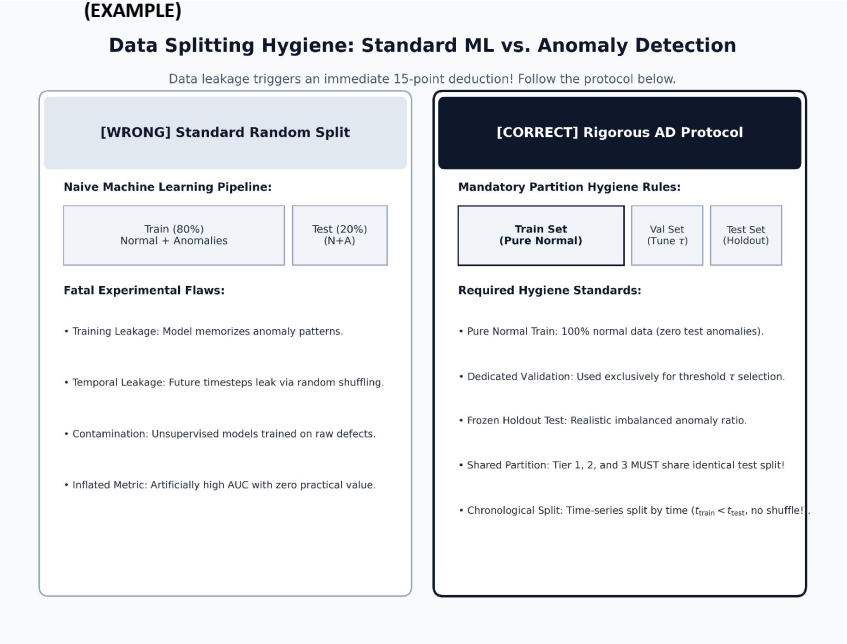
– Hyperparameters with search range and selected values; libraries; CPU / GPU.

Evaluation Metrics

– Choose at least two evaluation metrics that reflect your domain's cost of error (e.g., PR-AUC, ROC-AUC, F1 / Fβ, Precision@k, False Alarm Rate, Detection Delay).

– Justify your metric choices; explain what operational trade-offs they capture under your class distribution.

(EXAMPLE)



Common deductions Shuffled splits on time series; scaler fit on all data; test set reused for tuning; methods on different splits.

Main Results

Compare the three tiers under the protocol defined in §6.

8 pts
out of 100

Main Results Table

– One table: Tier 1, 2, 3 side by side on the identical test set across your chosen metrics (e.g., PR-AUC, ROC-AUC, F1, latency).

– Variation: report mean ± std over at least 3 seeds (or cross-validation folds); bold the best score in each column.

– Significance: test whether the gap to the strongest competitor is significant and name the test (e.g., paired t-test over seeds, bootstrap 95% CI of the metric gap (e.g., test PR-AUC or F1 gap)).

– Operational cost: report inference latency per sample (ms) and training time; peak memory is optional.

– Explain: why your method wins or loses against Tier 3, based on its assumptions; discuss the accuracy vs. cost trade-off.

(EXAMPLE) Main results (illustrative)

Method	PR-AUC	ROC-AUC	F1	ms / sample
T1 · 3σ on Amount	0.04 ± 0.00	0.61	0.07	0.01
T2 · Autoencoder	0.62 ± 0.03	0.95	0.64	0.08
T3 · XGBoost (sup.)	0.66 ± 0.02	0.97	0.70	0.05

Mean ± std over 5 seeds on the same test split. T2 vs. T3: paired t-test p = 0.08; bootstrap 95% CI of the PR-AUC gap [−0.09, 0.01] includes 0.

Reading: T2 is close to the supervised T3 without using fraud labels in training; the gap is not significant, so claim "comparable", not "better".

Common deductions Missing tier; methods on different splits; single run with no variation; gains claimed inside seed-to-seed variation.

Controlled Ablation

Isolate the value of each design choice by varying strictly one factor at a time.

10 pts
out of 100

Ablation Experiments

– Scope: at least 2 controlled ablations; at least 1 must target an anomaly-detection-specific choice.

– AD-specific topics: training contamination ratio (0/1/5/10%); scoring function (MSE vs. cosine vs. Mahalanobis); threshold strategy (percentile vs. best-F1 vs. budget); window size / sampling rate.

– Not sufficient alone: generic sweeps of learning rate, batch size, or dropout.

– Single variable: change one factor at a time; keep everything else fixed.

– Report: a table or curve with Δ vs. the full method, and whether Δ exceeds seed-to-seed std.

– Negative results: if a choice hurts, explain why; honest negative findings are valued.

(EXAMPLE) Training contamination ablation

Frauds in training set	PR-AUC	Δ vs. 0%
0% (normal-only, ours)	0.62 ± 0.03	—
1%	0.57 ± 0.03	−0.05
5%	0.44 ± 0.04	−0.18
10%	0.31 ± 0.05	−0.31

Reading: the autoencoder learns to reconstruct frauds it has seen, so their scores fall below τ; already at 1% the drop exceeds the seed std. Values are illustrative.

Common deductions No AD-specific ablation; settings differ in more than one factor; Δ reported without variation.

Error Analysis

Examine where the final model fails, and why.

8 pts
out of 100

Failure Case Studies

– Cases: at least 3 concrete test cases, including at least 1 false positive and 1 false negative.

– Numbers: sample ID, key raw feature values (or signal snippet / image patch), score s(x), and τ.

– Distance to τ: a near miss (boundary ambiguity) or a complete inversion?

– Cause: sensor noise, contextual shift, overlapping distributions, label noise, missing information.

– Shared pattern: what blind spot do the cases reveal, and what is the first fix you would try?

(EXAMPLE) Failure cases (τ = 0.042)

#	Type	Amount	s(x)	Likely cause
#1	FN	1.00	0.031	Card-testing fraud looks normal
#2	FN	99.99	0.040	Near miss just below τ
#3	FP	8,790	0.310	Rare but legitimate large purchase

Shared pattern: the model sees each transaction in isolation; small repeated test charges (#1) need per-card history features, not a bigger model. Values are illustrative.

Common deductions Fewer than 3 cases; no FP or no FN; cases without numbers; explanations not linked to features or data.

Practical Impact & Interpretability

Translate metrics into operational outcomes, and show why a sample was flagged.

8 pts
out of 100

Practical Impact

– Translate: what does your final metric difference mean for the stakeholder from §1?

– Different from §5: §5 asked how FP/FN costs shaped the design; this section asks what the final results mean in practice.

Interpretability

– Why flagged: explain at least one flagged sample.

– Evidence: SHAP values (tabular), per-feature reconstruction error (autoencoder), or pixel heatmaps (vision).

– Trust: would an analyst or engineer act on this explanation?

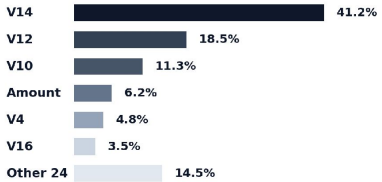
(EXAMPLE)

Section 8 · Case Study: Practical Impact & Interpretability

Explaining why sample #4,512 was flagged and translating metric deltas to operational savings

8.3 Sample #4,512 Feature-Wise Reconstruction Error Attribution

Score decomposition $s_j(x) = (x_j - \hat{x}_j)^2$ reveals which features triggered the alert:



Analyst Trust Assessment:

- High Confidence: V14 + V12 + V10 account for 71.0% of total error.
- Operational limitation: V-features are PCA components, so root-cause requires merchant code lookup.

8.2 Practical Operational Impact Translation

Translating abstract metric delta (+0.05 PR-AUC) into stakeholder labor & dollar outcomes:

Catch Rate @ 200 Budget	False Alarm Suppression	Operational Cost Impact
71 of 98 Frauds	**40 Fewer / Shift**	**$14,200 / Month**
T2 catches 72.4% vs 6.1% for Tier 1	Saves ~2.5 hrs analyst review/shift	Net fraud prevented minus customer friction

Common deductions Impact stated without numbers or users; explanation shown without interpreting it.

Conclusion & Future Work

Close the loop: answer the question posed in §1 with your key numbers.

3 pts
out of 100

Key Findings

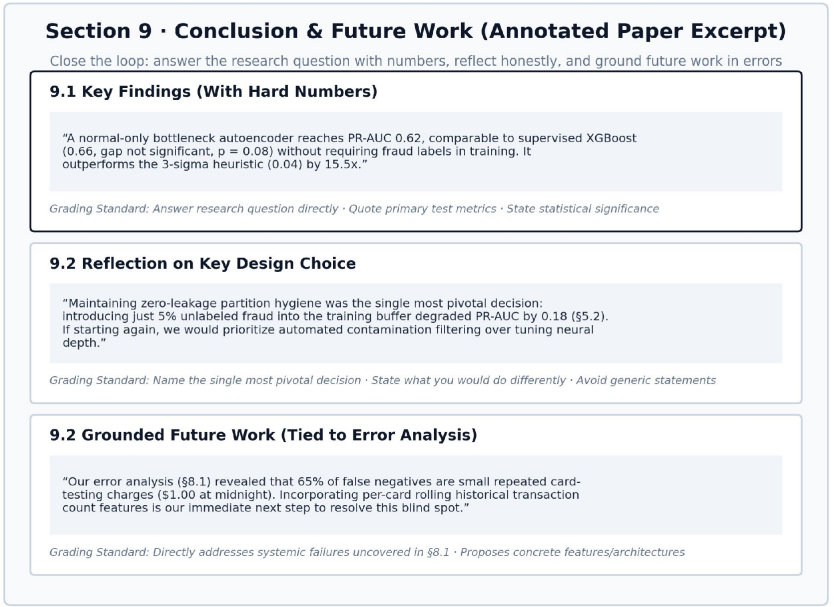
– Answer the research question from §1 in 2–3 sentences with your key numbers; introduce no new results.

Reflection on Key Design & Future Work

– Which decision mattered most? What would you do differently if you started again?

– 1–2 concrete next steps that address the failure patterns found in §8.

(EXAMPLE)



Common deductions Restating results without answering the research question; generic future work unrelated to §8.

Limitations & Deployment Considerations

State honestly what the evidence cannot show; admitting limitations is rewarded.

2 pts
out of 100

Limitations of the Study

– Data coverage: period, locations, population; are rare or new anomaly types missing?

– Assumptions: Gaussian scores, stationarity, clean normal training data — where do they break?

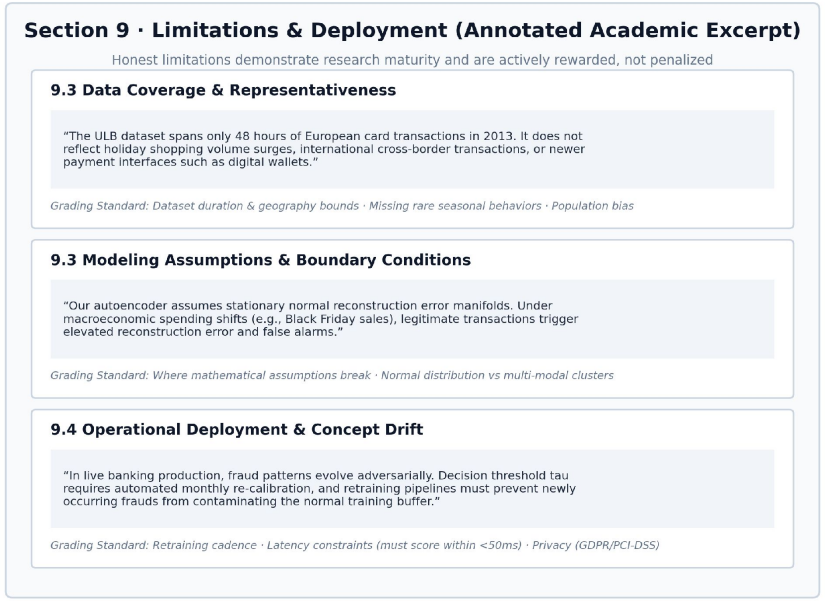
– Generalization: how far do the findings carry to other domains or larger data?

Deployment Considerations

– Concept drift: how often do normal patterns change, and when must the model be retrained?

– Latency & privacy: can inference meet the time window; which privacy or safety rules apply?

(EXAMPLE)



Common deductions No limitations discussed; limitations not linked to your own data, assumptions, or results.

Contributions of Each Member

List every member's concrete contributions as bullet points, and disclose how AI tools were used.

3 pts
out of 100

(EXAMPLE)

As it would appear in the report

Contributions of Each Member

· Member A (student ID): data collection, exploratory analysis & preprocessing; wrote §3–§4.
· Member B (student ID): Tier 2 method, pipeline, threshold study; wrote §5 and §7 (ablation).
· Member C (student ID): Tier 1/3 baselines, significance tests, error analysis; wrote §6, §7 (main table), §8.
· All members: presentation slides, conclusions, and proofreading.

AI Usage Statement

We used [tool name and version] to draft plotting code and polish wording. Method design, experiment design, and all conclusions were decided and verified by the team.

Requirements

– One bullet per member, naming concrete tasks and the sections or figures produced.

– AI usage: which tools, at which steps, and which decisions the team made and verified itself.

– Place this section before the references; it counts toward the 6 pages.

– Using AI tools is allowed; the team remains responsible for every result.

Free-riding? Teammates normally share one score, based on overall performance on each part. If contributions differ clearly, we may adjust individual scores accordingly. If a teammate does not contribute, contact the TAs.

PRESENTATION

Oral Presentation

Tell the story of your paper: a clear problem, evidence-based design choices, and honest results.

30%
of the HW2 grade

What to cover

These are suggested core elements, not a rigid slide sequence. Feel free to organize your presentation flow to tell your project's story best within 8 minutes.

- **Title & team**
- **Motivation** | what, for whom, and why
- **Problem formulation** | anomaly definition, inputs, imbalance
- **Data & key insight** | the plot that shaped your approach
- **Method** | pipeline, score $s(x)$, threshold τ
- **Challenges** | data issues, dead ends, surprises
- **Evaluation** | protocol, 3-tier results, ablation
- **Interpretation** | explanations, failure cases
- **Related work** | how your approach differs
- **Contributions** | who did what
- **Conclusion** | takeaways for the user

Date

Tue 2026-11-17, in class

Slides

PPTX or PDF, submitted with the report by 2026-11-16; present the submitted file

Time limit

Strictly 8 minutes per group

Graded on

Clarity of the story, evidence behind claims, readable figures, timing

SUBMISSION

E3 Submission

One submission per group on E3. Submission is due Mon 2026-11-16, 23:59.

Upload to E3: HW2_<TeamID>.zip

- **HW2_<TeamID>_report.pdf**
  - ACL template · At most 6 pages, excluding references.
  - GitHub repository link on the first page, below the title.
- **HW2_<TeamID>_presentation.pptx/.pdf**
  - The slides you present on 2026-11-17

Rules

- One member uploads for the whole group.
- Use the file names exactly as shown or there will be a deduction of 5 points for the HW2 paper report.
- If the GitHub repository link is not included on the first page of the paper report, the paper report will receive a score of 0.
- GitHub repository: your code, datasets and a short README.
- Pages after page 6 of the paper report are not read.

Mon 2026-11-16 23:59

HW2 Deadline

Report and slides on E3

Tue 2026-11-17

HW2 Presentation

In class

No late submissions
Files uploaded after 2026-11-16 23:59 are not graded.

GRADING

Grading Policy

HW2 grade = 70% report + 30% presentation + optional bonus.

70% Report

Scored out of 100 by section (see “Paper Structure”), then weighted 70%

30% Presentation

Story, evidence, clarity, timing, and Q&A

+6% Bonus

Up to 6% added to the HW2 grade via the LLM TA Judge (optional)

We assess whether

- Design decisions are justified
- Conclusions are supported by your own experiments
- Comparisons are fair (one shared, leakage-free split)
- Analyses cite concrete numbers and cases
- Writing and figures are clear
- The topic is creative and distinctive

LLM TA Judge

Automated reviewers score your report against the published rubric and return written feedback.

How the LLM TA Judge works →

Privacy & Anonymity (Zero-Data Retention)

- Endpoints: Your report goes through OpenRouter only to zero-data-retention (ZDR) endpoints (Azure, Google Vertex AI, Amazon Bedrock), which neither store it nor use it for training.
- Automatic Masking: Before sending, names, student IDs, emails, and GitHub accounts in text and page images are masked automatically.
- Draft Submission: Still submit an anonymized version with only your Team ID: compile with \usepackage[review]{acl} and write your Team ID as author. Do not upload confidential data.
- Email Privacy: Your email is used only to return feedback, never sent to AI models.

Have Fun!